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CE205
Databases and Information Retrieval

Anomalies, Inference Rules of Functional Dependencies, Keys and Attribute Closure

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Most of the slides based on

- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
- *Database Systems: Introduction to Databases to Data Warehouses* by Jukic, Vrbsky, Nestorov, sharma, 2nd Edition.

Learning Objectives

- Potential problems associated with redundant data in the base relations
- Concept of functional dependency, which describes the relationship between attributes
- Characteristics of functional dependencies
- How to identify functional dependencies for a given relation

If time permits

- What is closure of an attribute
- How functional dependencies identify the primary key for a relation.

Relational Database Model

- Relational database model - logical database model that represents a database as a collection of related tables
- Relational schema - visual depiction of the relational database model

- Relation - table in a relational database
  - A table containing rows and columns
  - Every relation is a table but every table is not a relation

Characteristics of Relations

- Rows contain data about an entity.
- Columns contain data about attributes of the entities.
- All entries in a column are of the same kind.
- Each column has a unique name.
- Cells of the table hold a single value.
- The order of the columns is unimportant.
- The order of the rows is unimportant.
- No two rows may be identical.

Example of a Non-relational Table

EmpID	EmpInfo	EmpInfo	EmpPhone	EmpBdate
0001	Joe	M	x-234	1/11/1995
0002	Sue	F	x-345	2/7/1993
0001	Joe	M	x-234	1/11/1995
0004	Pat	F	x-567, x-789	3/8/1981
0005	Mike	M	x-678	a long time ago

Multiple values in a Cell

Entries are not of the same kind

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Example of a Relational Table

EmployeeNumber	FirstName	LastName	Department	EmailAddress	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	518-834-1101
200	Mary	Abernathy	Finance	MA@somewhere.com	518-834-2101
300	Liz	Smathers	Finance	LS@somewhere.com	518-834-2102
400	Tom	Caruthers	Accounting	TC@somewhere.com	518-834-1102
500	Tom	Jackson	Production	TJ@somewhere.com	518-834-4101
600	Eleanore	Caldera	Legal	EC@somewhere.com	518-834-3101
700	Richard	Bandalone	Legal	RB@somewhere.com	518-834-3102

Alternative Terminology

- Although not all tables are relations, the terms table and relation are normally used interchangeably.
- The following sets of terms are equivalent:

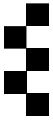
Table	**Column**	**Row**
Relation	**Attribute**	**Tuple**
File	**Field**	**Record**

Approaches to Design Relational Database Model

There are two main techniques for designing a database from system requirements. These are:

- Top-down Approach known as Entity-Relationship Modeling
  - *It starts by intuitively identifying objects of importance about which a system is to store data (entities) and then identifying the attributes which describe each of the entities and the relationships between the entities.*
- Bottom-Up approach known as Normalization.
  - *It starts with a collection of attributes and organises them into well-structured relations which are free from redundant data.*

Update Operations



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UPDATE OPERATIONS (1)

There are **three update operations**:

- **Insert operation**
  - Used for entering new data in the relation
- **Delete operation**
  - Used for removing data from the relation
- **Modify operation**
  - Used for changing the existing data in the relation

UPDATE OPERATIONS (2)

- Update operation terminology note: in practice there are two different uses of the term “update operation”:
  - A) Update operation as a collective term for insert, delete and modify operations
  - B) Update operation as a synonym for the modify operation
- we will use the term update operation as defined in A.

Insert operation

Before Insert

EMPLOYEE

EmpID	**EmpName**	**Salary**
1234	Becky	$75,000
2345	Molly	$50,000
3456	Rob	$55,000

After Insert

Inserting a record



EMPLOYEE

EmpID	**EmpName**	**Salary**
1234	Becky	$75,000
2345	Molly	$50,000
3456	Rob	$55,000
1324	Ted	$70,000

Delete OPERATION

Before Delete

EMPLOYEE

EmpID	**EmpName**	**Salary**
1234	Becky	$75,000
2345	Molly	$50,000
3456	Rob	$55,000

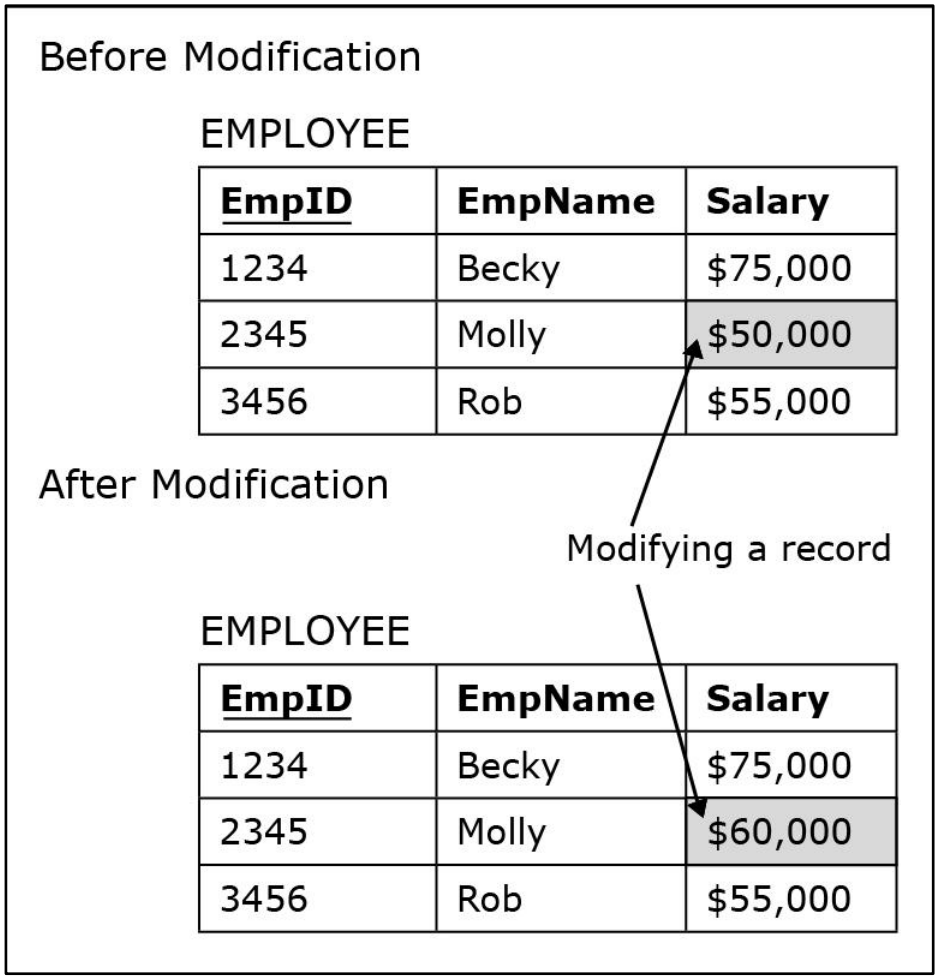
Deleting a record

After Delete

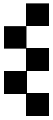
EMPLOYEE

EmpID	**EmpName**	**Salary**
1234	Becky	$75,000
3456	Rob	$55,000

Modify OPERATION



Data Redundancy and Update Anomalies



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Scenario for an Advertising Company:

- Design a database for an advertising company *"The Pressly Ad Agency"*. The company manages ad **campaigns** using **different methods** (each method is known as a *campaign modus*). Each campaign has a **starting date** and **duration** and is handled by a **person**.

Taken from *Database Systems: Introduction to Databases to Data Warehouses* by Jukic, Vebsky, Nestorov, sharme, 2nd Edition, Chapter 3.

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Assumptions

- Each modus has an identifier **ModusID** and it uses one media (TV, Radio, Print)

- Each ad campaign has
  - a unique identifier, a unique name, a start date, a duration,

  and

  - a campaign manager who has a name and a unique identifier

Relation AD CAMPAIGN MIX

Relation (AD CAMPAIGN MIX) with its records

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Data Redundancy

Data redundancy occurs when the same piece of data is stored in two or more separate places

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

The media and range values for campaign modus 1 repeated twice

The name of the campaign manager CM100 repeated three times

The name, start date, and duration of the campaign 222 repeated three times

Relations that contain redundant information may potentially suffer from update anomalies.

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UPDATE ANOMALIES

Update anomalies - anomalies in relations that contain redundant (unnecessarily repeating) data, caused by update operations

- **Insertion anomaly** - occurs when inserting data about one real-world entity requires inserting data about another real-world entity
- **Deletion anomaly** - occurs when deletion of data about a real-world entity forces deletion of data about another real-world entity
- **Modification anomaly** - occurs when, in order to modify one real-world value, the same modification has to be made multiple times

UPDATE ANOMALIES in Relation AD CAMPAIGN MIX

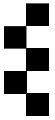
AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	**AutmnStyle20**	**6.9.2020**	**5 days**	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%
????	????	????	????	????	????	**7**	**Internet**	**National**	**????**

Deletion Anomaly Example:
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

Modification Anomaly Example:
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

Insertion Anomaly Example:
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.



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Problem

Find update anomalies in the relation **AIRPORT** (it captures the data about daily departing flights)?

Date	AirlineID	AirlineName	TerminalID	NumberOfGates	NumberOfDepartingFlights
11-Dec	UA	United	A	20	34
11-Dec	NW	Northwest	A	20	17
11-Dec	AA	American	A	20	11
11-Dec	DL	Delta	B	15	20
11-Dec	JB	Jet Blue	B	15	6
12-Dec	UA	United	A	20	29
12-Dec	DL	Delta	B	15	20
12-Dec	SWA	Southwest	C	15	17

Assumptions: (1) Each Airline has Unique Airline ID, Airline Name.
(2) Each terminal has unique Terminal ID, fixed NumberOfGates.
(3) Each airline has one permanently assigned terminal.
(4) Each Terminal can have multiple airlines.
(5) Each day flights depart from the airport for every airline.

Solution

Insertion anomaly:

- You cannot add a new terminal (D, 10) without adding an airline associated with it.

Deletion anomaly:

- If you were to delete airline (SWA, Southwest) the information about terminal C would also be removed.

Modification Anomaly:

- Changing the NumberOfGates value for terminal A from 20 to 25 would require making a change in four different records.